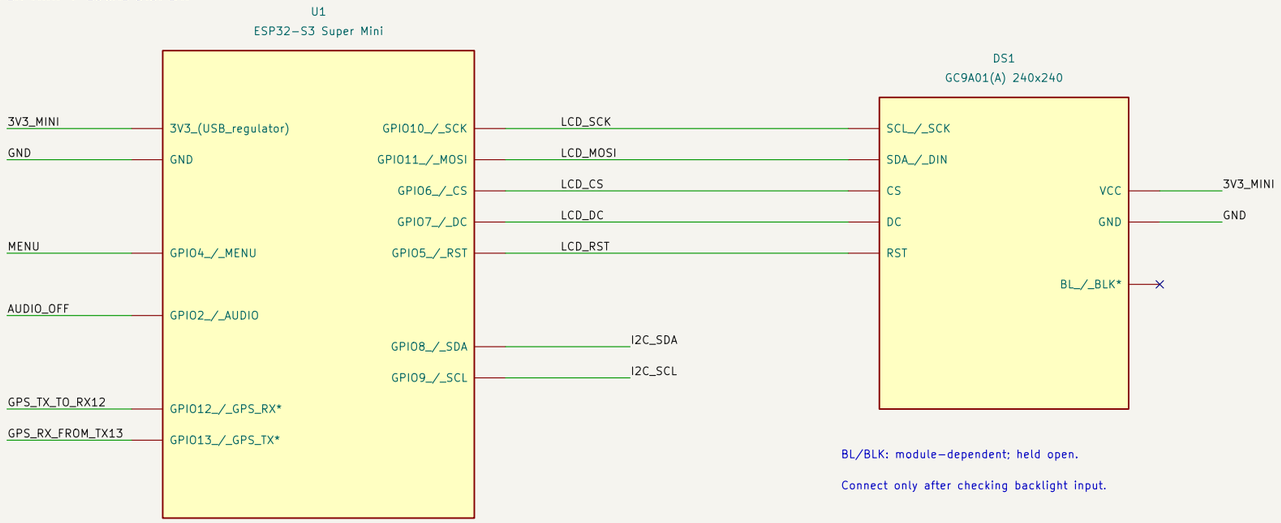
KU WATCH / MODULE INTERCONNECTION

REV A — REVIEW DRAFT | Native KiCad schematic | 16 SEP 2026

Solid green wires = electrical nets. Dashed lines = notes / planned links only. * = proposed or module-dependent.

01 SUPER MINI + LOCAL DISPLAY



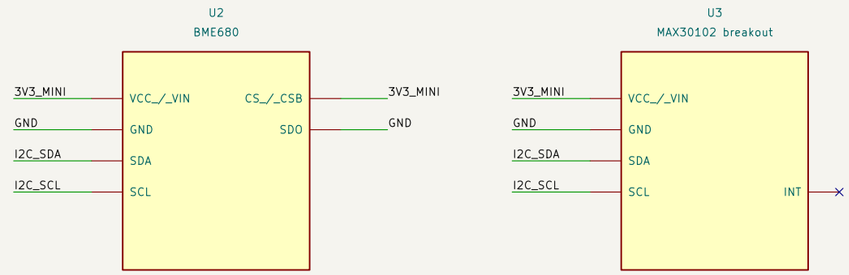
USB-C powers U1 for this draft.

Only used GPIOs shown; no SD module.

BL/BLK: module-dependent; held open.

Connect only after checking backlight input.

02 I2C SENSOR MODULES

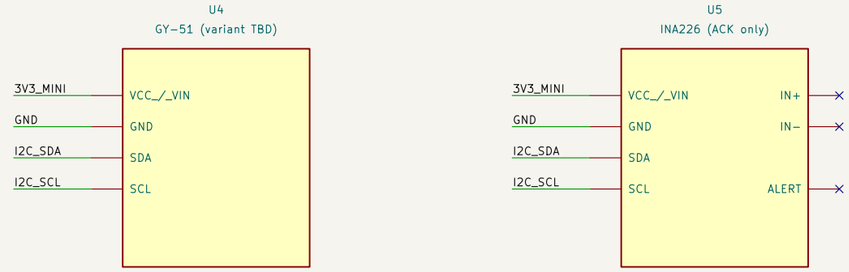


0x76 (CS high, SDO low)

Temp / humidity / pressure

0x57; IR + BPM

Regulated breakout, not bare IC.



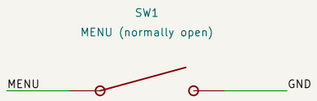
0x18 ACC + 0x1E MAG

Confirm chip / header markings.

0x40; sense path unassigned.

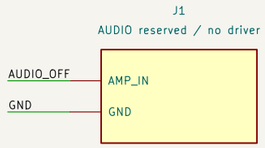
No voltage/current readout yet.

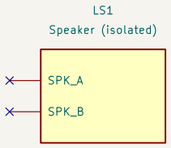
03 CONTROLS / GPS ON SUPER MINI



GPIO4 -> switch -> GND

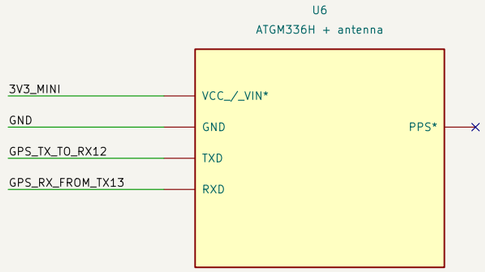
Internal pull-up in firmware.





GPIO2 disabled in firmware.

Amplifier/driver required.



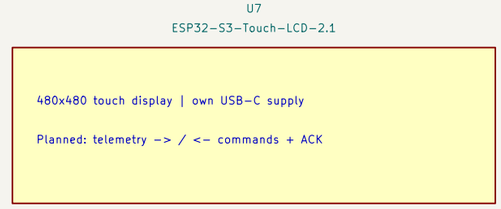
* Proposed UART: TXD -> GPIO12; RXD <- GPIO13.

Confirm exposed GPIOs, module VIN and header order.

Firmware UART/GPS not implemented. Antenna supplied.

04 WIRELESS REMOTE DISPLAY

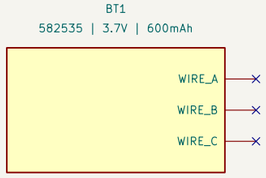
Super Mini < - - ESP-NOW TWO-WAY - - > Remote



No physical signal/ground wire between boards.

Same radio channel; ESP-NOW firmware pending.

05 BATTERY FOR SUPER MINI — POWER DESIGN PENDING



3-wire polarity/order unknown; third wire not identified.

REQUIRED POWER STAGE — PARTS NOT SELECTED

Charge / protection / regulated output / USB power sharing

Battery is isolated. Do not connect raw cell to 3V3.

BUILD NOTES

Module-level assembly draft, not a PCB.

Terminal IDs are functions, not header positions.

Verify 3.3V-compatible VIN / logic on breakouts.

I2C pull-ups: check installed module resistors.

No added pull-ups until bus loading is checked.

No footprints assigned. See README_TH.md.

GPIOs from firmware; GPS proposed; battery power stage pending

ESP32-S3 Super Mini + wireless touch display

Sheet: /
File: KU_Watch.kicad_sch

Title: KU Watch — module interconnection

Size: A3	Date: 2026-09-16	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1